

Qualitative data

Qualitative data, also called categorical data, describe categories.

Why is a course subject a category even if the school assigns it a number?

Quantitative data

Quantitative data, also called numerical data, record counts or measurements.

Does the number 201 mean the same kind of thing in “room 201” and “201 seats”?

Discrete data

Quantitative discrete data take separated, countable values, commonly from counting.

Can the number of open browser tabs be 4.5?

Continuous data

Quantitative continuous data measure quantities that can take any value within an interval.

A download lasts 3.2 seconds. Could a more precise timer report 3.24 seconds?

Continuous random variable

A continuous random variable has possible numerical values throughout an interval.

Could the unrounded time until a download completes be continuous?

Classifying a variable

A spreadsheet records each classroom's building name.

What type of data is this?

Classifying a count

A spreadsheet records the number of chairs in each classroom.

What type of data is this?

Classifying a measurement

A spreadsheet records each classroom's floor area in square meters.

What type of data is this?

Numbers as labels

Students have identification numbers such as 10482 and 10517.

Would the mean ID number describe a meaningful amount?

Numbers recoded as categories

A teacher records numerical quiz scores, then reports A, B, C, D, or F.

How does converting scores to categories change the information retained?

Bar graphs

A bar graph compares category counts or percentages using bar lengths.

For 12 blue, 8 black, and 5 red notebooks, what would the bars show?

Pie charts

A pie chart represents mutually exclusive parts of one whole.

Among 25 notebooks, 12 are blue, 8 black, and 5 red. What fraction belongs to each slice?

Pareto charts

A Pareto chart orders category bars from largest to smallest.

How would you arrange categories with counts 7, 19, 4, and 11?

Overlapping categories

Among 100 students, 65 use laptops and 55 use tablets. Some use both.

Would these percentages make a valid two-slice pie chart?

Missing responses

A class has 30 students. A device survey records 18 laptops and 9 tablets, with one response per student.

What is missing from this summary?

Comparing different totals

Synthetic surveys: Campus A has 40 tablet users among 100 respondents; Campus B has 60 among 200.

Which campus has the larger proportion of tablet users?

Simple random sampling

Every possible sample of the chosen size has the same chance of selection.

How could we choose 5 of 30 numbered students this way?

Random sampling

Random sampling uses chance to select population members. A simple random sample gives every group of the chosen size equal probability.

Does selecting every tenth ID after a random start make every possible group equally likely?

Random number selection

Choose 4 different IDs from 01 through 20.
Generated IDs: 07, 07, 23, 14, 02, 19.

Which IDs enter the sample?

Stratified sampling

Divide the population into groups called strata. In proportional stratified sampling, randomly select the same fraction from every stratum.

How could we sample 10% of students from each academic program?

Cluster sampling

Divide the population into groups called clusters. Randomly select clusters and include everyone in the selected groups.

What sample results if we randomly choose three class sections and survey everyone in them?

Systematic sampling

After a random start, select every k th member of a list.

On a list of 60 students, start at 4 and select every tenth student. Which IDs enter the sample?

Convenience sampling

A convenience sample uses readily available individuals.

What could go wrong if a campus survey uses only students in the nearest classroom?

Voluntary response

A voluntary-response or self-selected sample consists of people who choose to respond to an open invitation.

Who might respond to an online poll about parking fees?

Strata or clusters?

Plan A samples 5 students from every program. Plan B randomly chooses 2 programs and surveys everyone in them.

Which plan uses clusters?

Sampling with replacement

Sampling with replacement returns a selected member before the next selection, allowing repeats.

If a drawn numbered card is returned and shuffled in, can it appear again?

Sampling without replacement

Sampling without replacement allows each member to be selected only once.

How does leaving each drawn card out change the next selection?

Sampling variation

Different random samples can produce different statistics.

Two random samples report 42% and 47% support for a proposal. Must one be wrong?

Sampling error

Sampling error is the difference arising because a sample, rather than the whole population, is observed.

Can an unbiased random sample still give a statistic different from the population parameter?

Nonsampling error

Nonsampling errors arise from problems such as inaccurate measurement, data entry, nonresponse, or biased design.

Would recording 42 as 24 be sampling error or nonsampling error?

Sample size and precision

Larger random samples generally reduce sampling variability. They do not repair a biased selection procedure.

Which helps more: enlarging a biased poll or correcting its selection method?

Variation in data

Observations within one sample can differ.

Three printers take 9, 12, and 18 seconds for the same file. What might account for the variation?

Sampling bias

Sampling bias occurs when selection systematically favors some population members or outcomes.

Would surveying 10,000 library visitors remove bias in a survey of all students?

Nonresponse

Nonresponse occurs when selected individuals do not provide data. A college randomly invites 400 students; only 35 reply.

What concern remains even though the invitations were random?

Question wording

“Should the college waste money on longer library hours?”

How could we ask about the same issue more neutrally?

Undue influence

Undue influence occurs when collection conditions or wording pressure people toward a response.

How could asking about a service while its manager watches affect answers?

Causality and confounding

Association means variables vary together. Causality means one influences the other; confounding makes their effects difficult to separate.

Do higher scores among voluntary tutoring users prove tutoring caused the difference?

Self-funded studies

A self-funded or self-interest study is supported by someone with a stake in its result.

What methods and disclosures would you examine before accepting a software vendor's study?

A sampling choice

A college wants feedback from both online and on-campus students.

What sampling plan would you propose?